particles—the symmetry between them means there was symmetry between the forces. At high enough temperatures, therefore, temperatures that would vaporize today's Higgs-filled vacuum, there is no distinction between the weak nuclear force and the electromagnetic force. At high enough temperatures, that is, the Higgs ocean evaporates; as it does, the distinction between the weak and electromagnetic forces evaporates, too.

Glashow, Weinberg, and Salam had extended Maxwell's century-old discovery by showing that the electromagnetic and weak nuclear forces are actually part of one and the same force. They had *unified* the description of these two forces in what is now called the *electroweak* force.

The symmetry between the electromagnetic and weak forces is not apparent today because as the universe cooled, the Higgs ocean formed, and—this is vital—photons and W and Z particles interact with the condensed Higgs field differently. Photons zip through the Higgs ocean as easily as B-movie has-beens slip through the paparazzi, and therefore remain massless. W and Z particles, though, like Bill Clinton and Madonna, have to slog their way through, acquiring masses that are 86 and 97 times that of a proton, respectively. (Note: this metaphor is not to scale.) That's why the electromagnetic and weak nuclear forces appear so different in the world around us. The underlying symmetry between them is "broken," or obscured, by the Higgs ocean.

This is a truly breathtaking result. Two forces that look very different at today's temperatures—the electromagnetic force responsible for light, electricity, and magnetic attraction, and the weak nuclear force responsible for radioactive decay—are fundamentally part of the same force, and appear to be different only because the nonzero Higgs field obscures the symmetry between them. Thus, what we normally think of as empty space—the vacuum, nothingness—plays a central role in making things in the world appear as they do. Only by vaporizing the vacuum, by raising the temperature high enough so that the Higgs field evaporated—that is, acquired an average value of zero throughout space—would the full symmetry underlying nature's laws be made apparent.